

Manish

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EDUCATION

Indian Institute of Technology, Kharagpur

Bachelor of Technology in Aerospace Engineering

West Bengal, India

Aug 2018 – Jun 2023

EXPERIENCE

Senior Data Scientist

TCG Digital

Sept 2025 – Present

New Delhi, IN (Hybrid)

- **CataLLM (Agentic AI & Research):** Spearheaded the development of an AI discovery engine that automates catalyst formulation by analyzing **50k+ global patents**; utilized fine-tuned LLMs to generate data-backed hypotheses for material characteristics, achieving a **65% relevance rate** via custom scoring on **GCP**.
- **Intrusion Detection:** Designed and deployed an AI-driven multi-modal system for the pipelines division, reducing false positives by **40%** and enhancing anomaly detection accuracy by **30%**.
- **Automated Proposal Generation:** Implemented private LLM-based automation for technical proposal generation, enabling structured parsing and reasoning from complex PDF documents to streamline business workflows.

Data Scientist

TCG Digital

Aug 2023 – Sept 2025

Kolkata, IN (Hybrid)

- **Oil and Gas Industry :** Engineered **hybrid optimization** using ML prediction and differential evaluation algorithm to optimize ethylene production in oil and gas operations and generating **\$ 2M** annual savings
- **Textile Industry :** Spearheaded the implementation of an advanced custom **margin maximization** algorithm, reducing scheduling and capacity planning time **30 %** and enhancing resource utilization across facilities
- **Petrochemical Industry :** Applied advanced data science techniques to create **98 % accurate** location specific price forecasting model along with transportation cost, providing **3-month** market insights using **xgboost model**
- **Offshore Drilling:** Developed ML models to predict drilling fluid rheology, scaling to **1M+ records** across **50+ global locations** with automated pipelines and **87 %** accuracy tracking for the world's largest offshore drilling company.

Quantitative Analyst Intern

Z3 Systems

Aug 2022 – Jul 2023

California, USA (Remote)

- **Dashboard :** Engineered a market analytics dashboard monitoring **500+** metrics for US Stocks, ETFs and MFs
- **Stock Selection :** Developed quantitative screening algorithms for identifying optimal stock entry points
- **Bear bottom Analysis :** Conducted Analysis on **10,000+** stocks and Mutual Funds Historical Data , identifying patterns that increased portfolio resilience by **18 %** and improved recovery time by **25 %** during bear markets

Data Science Intern

Anagatam Technologies

Apr 2022 – Jun 2022

Bangalore, IN

- Automated financial data extraction from annual reports using Python, eliminating manual analysis time by **30 %**
- Built stock forecasting model using **LSTM & RNN** algorithm with **97.66 %** accuracy for 7-day price prediction
- Implemented **NLP**-based sentiment analysis system correlating chairman's messages with stock performance trend

PROJECTS

InHouseLLM Project | Generative AI

Aug 2024 – Oct 2024

- Led the InHouseLLM Initiative, developing a bespoke Language Model system with FastAPI for streamlined APIs
- Ensured privacy in parsing and embedding with **Docker** containers, while scaling application using **Elasticsearch**
- Showcased the effectiveness of the system with response time under **100 milliseconds**, serving **1,000** users

TECHNICAL SKILLS

Languages: Python | *SQL* | *C++* | *MATLAB* | *C* | *JavaScript* | *HTML* | *CSS*

Machine Learning & AI: LLMs | *LangChain* | *RAG* | *Prompt Engineering* | *Vector Database* | *OpenAI API*

MLOps & Deployment: Docker | *FastAPI* | *MLflow* | *Airflow* | *CI/CD* | *Kubernetes*

Cloud Platforms : AWS | *Azure* | *GCP*

Visualization & Tools : Power BI | *Tableau* | *Streamlit* | *Git* | *ElasticSearch*